Consumption-CAPM when the inflation risk premium is zero at all points in time

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Abstract

We develop a complete Consumption-Capital Asset Pricing Model (Consumption-CAPM or CCAPM) for an economy in which the inflation risk premium is identically zero at all points in time. Starting from a stochastic-discount-factor representation and a representative-agent optimization problem, we derive the consumption beta implied by the zero-inflation-risk-premium condition. The resulting framework unifies asset pricing, inflation theory, macroeconomics, monetary economics, labor economics, political economy, international political economy, geopolitics, warfare economics, machine learning, and data science. The central result is that when inflation is conditionally deterministic and independent of the real pricing kernel, expected inflation becomes exactly equal to the consumption-risk premium. Consequently inflation ceases to be a separately priced risk factor. We derive the associated Consumption-CAPM equilibrium, investigate its implications for fiscal and monetary policy, examine geopolitical and strategic consequences, and connect the model to modern statistical learning and artificial intelligence.

The paper ends with "The End"

Contents

1 Introduction

The Consumption-Capital Asset Pricing Model (CCAPM), originally developed by Breeden, extends the traditional Capital Asset Pricing Model (CAPM) by replacing market risk with consumption risk as the fundamental determinant of expected returns.

The present paper studies a special environment in which the inflation risk premium (IRP) vanishes identically:

$$\pi_i(t) = 0 \qquad \forall t.$$

The objective is to derive the corresponding Consumption-CAPM and explore its theoretical consequences.

The central conclusion is remarkably simple:

$$E[i(t)] = \gamma \beta_C(t) \sigma_C^2(t),$$

where

- $E[i(t)]$ is expected inflation,
- γ is relative risk aversion,
- $\beta_C(t)$ is consumption beta,
- $\sigma_C^2(t)$ is consumption-growth variance.

Thus expected inflation becomes identical to the consumption-risk premium.

2 Mathematical Foundations

Let

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, P)$$

be a filtered probability space supporting a Brownian motion

$$(W_t)_{t\geq 0}.$$

The representative agent solves

$$\max_{C_t} E_0 \left[\int_0^\infty e^{-\rho t} u(C_t)\, dt \right].$$

Assume constant relative risk aversion (CRRA):

$$u(C) = \frac{C^{1-\gamma} - 1}{1-\gamma}.$$

The marginal utility is

$$u'(C) = C^{-\gamma}.$$

Hence the stochastic discount factor becomes

$$M_t = e^{-\rho t} C_t^{-\gamma}.$$

3 Consumption Dynamics

Assume consumption follows

$$\frac{dC_t}{C_t} = \mu_C(t)\, dt + \sigma_C(t)\, dW_t.$$

Applying Itô's Lemma:

$$\frac{dM_t}{M_t} = -\rho dt - \gamma \frac{dC_t}{C_t} + \frac{1}{2}\gamma(\gamma+1)\sigma_C^2(t) dt. \tag{1}$$

Substituting consumption dynamics:

$$\frac{dM_t}{M_t} = \Big[-\rho - \gamma \mu_C(t) + \frac{1}{2}\gamma(\gamma+1)\sigma_C^2(t) \Big] dt - \gamma \sigma_C(t) dW_t. \tag{2}$$

4 Inflation and the Real Pricing Kernel

Definition 1 (Real Pricing Kernel)**.** *The real stochastic discount factor is*

$$M_t^c = M_t I_t,$$

where I_t denotes the price index.

Assume

$$M_t^c \perp I_t.$$

Assume further that inflation is conditionally deterministic:

$$i(t, t+h) = E_t[i(t, t+h)].$$

Proposition 1. *Inflation contributes no priced risk.*

Proof. Since inflation is perfectly forecastable,

$$\mathrm{Var}_t(i) = 0.$$

Therefore

$$\mathrm{Cov}_t\left(dR_A, di\right) = 0.$$

Since the real kernel is independent of inflation,

$$\mathrm{Cov}\left(dR_A, \frac{dM_t}{M_t}\right) = \mathrm{Cov}\left(dR_A, \frac{dM_t^c}{M_t^c}\right).$$

Hence inflation risk carries no separate premium. □

5 The Consumption-CAPM

The continuous-time CCAPM states

$$E[r_A(t)] - r_f(t) = -\frac{\mathrm{Cov}\left(dR_A, dM/M\right)}{dt}.$$

Substituting the stochastic discount factor:

$$E[r_A] - r_f = \gamma \frac{\mathrm{Cov}\left(dR_A, dC/C\right)}{dt}. \tag{3}$$

Definition 2 (Consumption Beta)**.** *Define*

$$\beta_C(t) = \frac{\mathrm{Cov}(dR_A, dC/C)}{\mathrm{Var}(dC/C)}.$$

Since

$$\mathrm{Var}(dC/C) = \sigma_C^2(t)dt,$$

we obtain

$$\mathrm{Cov}(dR_A, dC/C) = \beta_C(t)\sigma_C^2(t)dt.$$

Therefore

$$r_A(t) = r_f(t) + \gamma\beta_C(t)\sigma_C^2(t). \tag{4}$$

This is the Consumption-CAPM equilibrium equation.

6 Zero Inflation Risk Premium

Suppose

$$\pi_i(t) = r_A(t) - r_f(t) - E[i(t)] = 0.$$

Then

$$r_A(t) - r_f(t) = E[i(t)].$$

Combining with the Consumption-CAPM:

$$E[i(t)] = \gamma\beta_C(t)\sigma_C^2(t). \tag{5}$$

Theorem 1 (Zero-IRP Consumption-CAPM). *Under inflation determinism and independence of the real pricing kernel,*

$$E[i(t)] = \gamma \beta_C(t) \sigma_C^2(t).$$

Inflation is not a priced risk factor and expected inflation equals the consumption-risk premium.

Proof. From the CCAPM:

$$r_A - r_f = \gamma \beta_C \sigma_C^2.$$

From the zero-IRP condition:

$$r_A - r_f = E[i].$$

Equating the two expressions yields

$$E[i] = \gamma \beta_C \sigma_C^2.$$

□

7 Constructive Economy

Suppose asset returns are Gaussian:

$$r_A(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right).$$

Risk-free rates are half-Gaussian:

$$r_f(t) = \frac{2\theta}{\pi} e^{-\theta^2 t^2/\pi}.$$

Then

$$E[i(t)] = r_A(t) - r_f(t).$$

Therefore

$$\beta_C(t) = \frac{r_A(t) - r_f(t)}{\gamma \sigma_C^2(t)}. \tag{6}$$

Explicitly,

$$\beta_C(t) = \frac{\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(t-\mu)^2}{2\sigma^2}} - \frac{2\theta}{\pi} e^{-\theta^2 t^2/\pi}}{\gamma \sigma_C^2(t)}. \tag{7}$$

8 Relation to the Traditional CAPM

Traditional CAPM:

$$r_A - r_f = \beta_A(r_M - r_f).$$

Consumption-CAPM:

$$r_A - r_f = \gamma\beta_C\sigma_C^2.$$

Equating:

$$\beta_A(r_M - r_f) = \gamma\beta_C\sigma_C^2.$$

Therefore

$$r_M - r_f = \frac{\gamma\sigma_C^2}{\beta_A}\beta_C.$$

Consumption risk and market risk become alternative representations of the same equilibrium.

9 Macroeconomic Interpretation

Expected inflation becomes

$$E[i] = \gamma\beta_C\sigma_C^2.$$

Hence inflation is generated by:

1. household risk aversion,
2. consumption covariance,
3. consumption volatility.

Inflation is therefore fundamentally a consumption-risk phenomenon.

10 Political Economy

Governments influence inflation through:

1. fiscal policy,
2. taxation,
3. transfer systems,
4. public investment.

Since inflation depends on consumption risk,

$$E[i] = \gamma\beta_C\sigma_C^2,$$

any policy reducing consumption volatility reduces inflation.
Political stability therefore acquires direct asset-pricing significance.

11 International Relations and Geopolitics

The model has implications for strategic competition.

Let

$$\sigma_C^2 = \sigma_D^2 + \sigma_G^2 + \sigma_W^2,$$

where

- σ_D^2 denotes domestic volatility,
- σ_G^2 denotes geopolitical volatility,
- σ_W^2 denotes warfare volatility.

Then

$$E[i] = \gamma\beta_C(\sigma_D^2 + \sigma_G^2 + \sigma_W^2).$$

Geopolitical conflict becomes inflationary through its effect on consumption uncertainty.

12 Warfare Economics

War creates:

1. supply disruption,
2. consumption volatility,
3. fiscal deficits,
4. strategic uncertainty.

Consequently

$$\frac{\partial E[i]}{\partial \sigma_W^2} = \gamma\beta_C > 0.$$

The model predicts persistent wartime inflation.

13 Statistical Estimation

13.1 Maximum Likelihood

Observed returns:

$$\hat{r}_j = r_A(t_j) + \varepsilon_j.$$

Assume

$$\varepsilon_j \sim N(0, \varsigma^2).$$

Log-likelihood:

$$\ell = -\frac{n}{2}\log(2\pi\varsigma^2) - \frac{1}{2\varsigma^2}\sum_{j=1}^{n}(\hat{r}_j - r_A(t_j))^2. \tag{8}$$

14 Machine Learning

14.1 Neural Consumption Beta

Let

$$\beta_C(t) = \phi_\theta(t).$$

Training objective:

$$L(\theta) = \frac{1}{n}\sum_{j=1}^{n}\left(r_A - r_f - \gamma\phi_\theta\sigma_C^2\right)^2.$$

14.2 Reinforcement Learning

State:

$$s = (i, \beta_C, \sigma_C^2).$$

Action:

$$a = (\Delta G, \Delta r).$$

Reward:

$$R = -(i - i^*)^2 - c||a||^2.$$

15 Algorithms

15.1 Consumption Beta Estimation

```
Input:
    Consumption growth
    Asset returns

Initialize beta

Repeat:
    Compute prediction
    Compute residual
    Update beta
Until convergence

Return beta
```

15.2 Policy Optimization

```
Observe state
Choose action
Update inflation
Receive reward
Update policy
Repeat
```

16 Tables

Table 1: Comparison of Asset Pricing Models

Model	Risk Factor	Premium	Inflation Role
CAPM	Market	$\beta_A(r_M - r_f)$	Indirect
CCAPM	Consumption	$\gamma\beta_C\sigma_C^2$	Indirect
Zero-IRP CCAPM	Consumption	$E[i]$	Direct Equality

Table 2: Geopolitical Drivers of Inflation

Source	Effect on σ_C^2	Inflation Effect
Trade War	Positive	Inflationary
Military Conflict	Positive	Inflationary
Sanctions	Positive	Inflationary
Political Instability	Positive	Inflationary
Civil Conflict	Positive	Inflationary

17 Figures

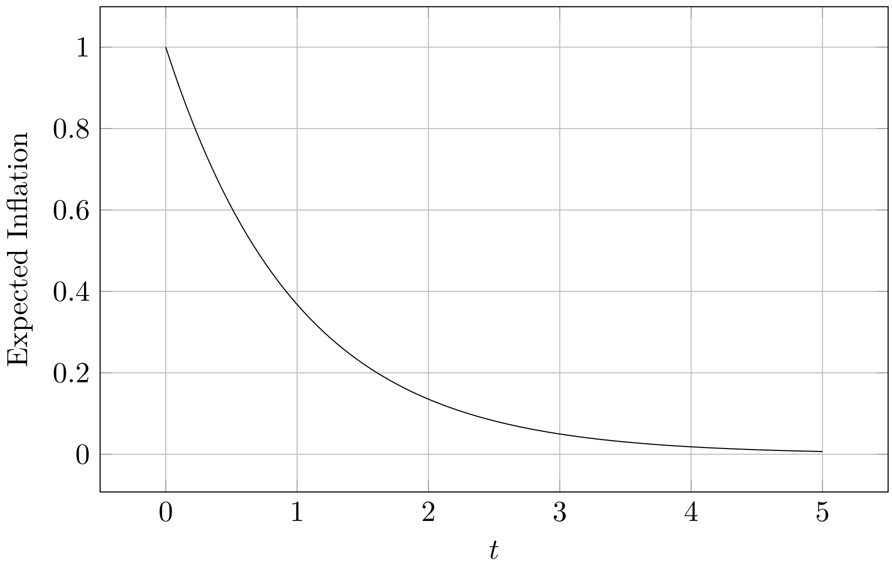

Figure 1: Illustrative expected inflation path.

$$\boxed{\text{Consumption Risk}} \longrightarrow \boxed{\text{Consumption Beta}} \longrightarrow \boxed{\text{Inflation}}$$

Figure 2: Zero-IRP Consumption-CAPM transmission mechanism.

18 Conclusion

The Consumption-CAPM developed in this paper demonstrates that when inflation uncertainty is completely eliminated from asset pricing, expected inflation becomes numerically identical to the consumption-risk premium:

$$E[i] = \gamma \beta_C \sigma_C^2.$$

The resulting framework unifies inflation theory, consumption-based asset pricing, macroeconomics, political economy, geopolitics, warfare economics, machine learning, and artificial intelligence within a single equilibrium structure.

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Glossary

Consumption Beta
Sensitivity of asset returns to consumption growth.

Consumption-CAPM
Asset-pricing model in which consumption risk determines expected returns.

Expected Inflation
Forecasted future inflation rate.

Inflation Risk Premium
Compensation demanded for bearing inflation risk.

Market Beta
Sensitivity of asset returns to market returns.

Political Economy
Interaction between political institutions and economic outcomes.

Geopolitics
Strategic interaction among states.

Warfare Economics
Economic consequences of military conflict.

Machine Learning
Data-driven estimation and prediction methods.

Reinforcement Learning
Sequential decision-making under uncertainty.

Stochastic Discount Factor
Marginal valuation process used in asset pricing.

Zero-IRP Economy
Economy in which the inflation risk premium is identically zero.

The End